

A cascaded force/position hybrid control method based on ultrasound robots

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Abstract— In the ultrasound robot automatic scanning, the robot needs to maintain a stable contact force in the scanning normal direction to ensure image quality, while maintaining a safe human-computer interaction force in other directions. Therefore, achieving stable and safe contact force control and accurate position tracking has become a key challenge in complex environments. Therefore, this paper proposes a cascaded force/position hybrid control method to overcome the problem that it is difficult for traditional force/position hybrid control to balance accuracy and safe interaction in dynamic environments, especially the risk of force control failure in scenarios such as sudden shedding. Firstly, the median filter is dynamically compensated to suppress the noise of the force sensor in real time and eliminate the interference of the Coriolis force to improve the performance of the force signal. Then, the task space is decomposed into tangential and normal directions, Cartesian impedance control is used in the tangential direction to achieve compliant position tracking, and the cascaded force control strategy of impedance feedforward and PID feedback is fused in the normal direction to improve the force control performance. In addition, strict null-space projection is introduced to optimize redundant degrees of freedom and significantly reduce joint fluctuations. The abdominal phantom force tracking experiment shows that compared with the pure PID control, the proposed method reduces the standard error of the mean force by 24.53%, the fluctuation is smaller, the maximum contact force is reduced by 4.15%, the interaction safety is higher, and the impedance compensation force can effectively deal with the sudden detachment scene. Finally, high-precision force tracking is achieved in planar, curved and abdominal phantom scanning, which shows the effectiveness and reliability of the method.

Keywords—ultrasound robot, Cartesian impedance, force control, redundancy optimization

I. INTRODUCTION

Ultrasonography has become a widely used diagnostic method in clinical practice because it is non-invasive, radiation-free, easy to operate, and low cost [1]. Traditional ultrasonography relies on manual operation by physicians, but due to the shortage of resources and inexperience of physicians, it is difficult to cope with the increasing demand of patients, resulting in difficulty

in ensuring the efficiency and accuracy of examinations [2]. Robots have the characteristics of high precision, strong consistency, and long-term stable operation, which can not only significantly improve the accuracy and repeatability of ultrasound examinations, but also effectively alleviate the pressure on medical resources, so ultrasound robots have become an important force to promote the intelligent upgrading of the field of ultrasound diagnosis [3].

Ultrasound robots need to be in contact with the patient for long periods of time while accurately replicating the doctor's scanning movements. In this process, the robot dynamically adjusts the contact force through compliance control technology to maximize human safety. At the same time, the contact conditions of the ultrasound probe also have a significant impact on the quality of the obtained images. Therefore, maintaining safe and stable contact between the ultrasound probe and human tissues and precise force control are the key issues of automated ultrasound robots [4]. To this end, researchers have used a variety of control strategies to address this challenge. Allard et al. [5] used a position-controlled approach that relied on precise robotic arm movements to maintain the required contact force, providing a straightforward idea for force regulation. However, this method is highly dependent on precise motion control, and has poor adaptability to changes in environment and object characteristics, making it difficult to ensure stability and safety in complex interactions such as soft tissue scanning. Fuerst et al. [6] used compliance control to maintain a constant expectation force within a specified range, allowing the system to adapt to different tissue characteristics while ensuring patient safety and imaging accuracy. However, this strategy usually presets a fixed compliance characteristic, has limited dynamic adjustment ability, and lacks flexibility in scenarios that require rapid response or complex force interaction. Jiang et al. [7,8] used impedance control to dynamically adjust the robot's compliance based on the measured force to achieve safer, consistent contact, with the advantage of active dynamic response. However, impedance control essentially functions as a displacement source. That is, it adjusts the output force based on force feedback after specifying a desired position or trajectory. This approach can be limited in scenarios requiring precise contact force control, such as maintaining a specific pressure, and is sensitive to force sensor noise. As a complementary approach, admittance control takes a different approach [9]: it converts the error

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between the measured desired force and the actual force into a speed or position correction command, and indirectly controls the contact force by adjusting the robot's trajectory. This ability to generate motion instructions based on force errors gives it a potential advantage in applications that require high-precision force tracking, allowing for more straightforward implementation of the target force. For example, Fang B [10] used an admittance force control algorithm on a cooperatively controlled six-degree-of-freedom robotic ultrasound system, which enabled the robotic arm to comply with the operator's movements and assist in the precise application of force during the scanning process, effectively reducing the force actively exerted by the doctor. However, since admittance control is to control the force indirectly by correcting the motion, there is an essential contradiction between position control and force control when it is necessary to accurately track the desired trajectory at the same time, such as a specific scan path and maintain the desired contact force. Kim et al. [11,12,13] employ a hybrid force/position control to decompose the task space into a force-controlled subspace and a position-controlled subspace, aiming to dynamically adjust the probe position and the applied force independently. This method performs well in specific structured tasks, but it has shortcomings in ensuring the accuracy of force control and the safety of environmental interaction. In summary, although many achievements have been made in the existing research on the safety interaction force control of ultrasonic robots, the main difficulty is that ultrasound robots need to be able to maintain stable contact forces in the direction of the scan normal to get accurate ultrasound images in an unknown scanning environment, but also need to maintain safe interaction forces in other directions. In addition, the existing methods still have limitations in taking into account the image quality and safety interaction, especially in the interaction failure scenarios such as accidental probe slippage, and the protection mechanism of the system is insufficient.

In this paper, we propose a cascaded force/position hybrid control method for a 7-DOF ultrasound robot with redundancy optimization. Firstly, the median filter is dynamically compensated to suppress the noise of the force sensor in real time, and the Coriolis force interference is eliminated to improve the performance of the force signal. Secondly, the Cartesian impedance control is implemented in the tangential direction of the ultrasonic robot, so that the system realizes a flexible and safe position control effect in the tangential direction, thereby improving the overall control performance, and effectively inhibiting the abrupt movement of the manipulator through dynamic stiffness adjustment. In the scanning normal, a cascaded force control strategy combining impedance feedforward compensation and PID feedback control is used to maintain a stable contact force between the ultrasound probe and human tissue. Finally, strict null-space projection is introduced to optimize redundant degrees of freedom to significantly reduce joint fluctuations.

II. CASCADED FORCE/POSITION HYBRID CONTROL BASED ON ULTRASOUND ROBOTS

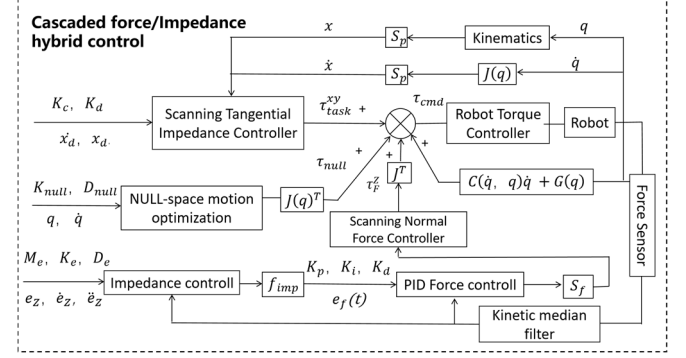


Fig. 1. Cascade force/position hybrid control of a 7-degree-of-freedom ultrasonic robot based on redundancy optimization

Fig. 1 shows the cascaded force/position hybrid control architecture of a 7-degree-of-freedom ultrasonic robot based on redundancy optimization, which aims to improve the robustness and accuracy of the control. The framework uses a dynamic compensation median filter in the signal processing layer to suppress the noise of the force sensor in real time, and eliminate dynamic interference such as Coriolis force, so as to provide high-quality force signals for subsequent control. The task decomposition layer decouples the task space into tangential and normal directions, the tangential direction is controlled by Cartesian impedance to achieve compliant position tracking, and the normal direction is fused with impedance feedforward and PID feedback control to enhance the force control performance. The redundancy optimization layer uses a strict null-space projection algorithm to optimize the redundant degrees of freedom and effectively inhibit joint fluctuations. The processing results of each level are finally converged to the torque controller, which drives the robot to achieve high-performance compliant interactive control. This study focuses on the ultrasonic task control of the high-precision Franka force-controlled robotic arm, which itself has good force control accuracy. The robot adopts a high-precision joint torque controller to achieve precise control, and the control torque τ_{cmd} output by the algorithm to the robot joint torque controller is as follows:

$$\tau_{cmd} = \tau_{task}^{xy} + \tau_F^z + \tau_{null} + C(\dot{q}, q)\dot{q} + G(q) \quad (1)$$

where τ_{task}^{xy} is the joint torque input corresponding to the scanning normal force controller, τ_F^z is the joint torque input corresponding to the ultrasonic robot scanning tangential impedance controller, τ_{null} is the null- space control input, $C(\dot{q}, q)\dot{q}$ is the vector of the robot's Coriolis force and centrifugal force, and $G(q)$ is the gravity vector.

A. Kinetic compensation median filter

In the ultrasonic robot contact force control, the force/position hybrid control can directly act on the force and position of the manipulator, but the measured force is often not accurate, the gravity and inertial force interfere greatly, the measurement noise is large, the force control of the manipulator has a large error with the expected value, the application is limited and the requirements for the force sensor are high. In order to solve the above problems, the controller

innovatively introduces the median filtering mechanism of dynamic compensation. In the control process, in order to eliminate the impulse noise and occasional peak interference generated by the robot external force sensor during contact interaction, in this study, the sliding window median filter is applied independently to each component of the six-dimensional measurement vector $F_{\text{meas}}(n) \in \mathbb{R}^6$ in each control cycle. Let the window length N be an odd number (if the input is an even number, it will automatically degenerate to no filtering), by maintaining a buffer array of length N $m_i(n) = [m_0(n), \dots, m_{N-1}(n)]$ to achieve sliding update, and sort the contents of the array in ascending order and take the median value as the filtered output:

$$\tilde{F}_i(n) = S_{N+1/2} \quad (2)$$

The median operation has the characteristics of nonlinear removal of extreme values, which can effectively suppress single-point noise and occasional pulses while retaining the real force mutation. Furthermore, in order to eliminate the interference of Coriolis force and inertial force caused by motion, the Jacobian matrix $J(q(n))$ and Coriolis vector $C(q(n), \dot{q}(n))$ are calculated by using the current joint angle $q(n)$ and velocity $\dot{q}(n)$ to obtain the compensating force:

$$F_{\text{cor}}(n) = J(q(n))C(q(n), \dot{q}(n)) \quad (3)$$

The final external forces are estimated to be the difference between the two:

$$\hat{F}(n) = \tilde{F}(n) - F_{\text{cor}}(n) \quad (4)$$

At this time, the dynamic disturbance of the system itself has been removed, and the impact of measurement noise on the force control loop has been eliminated, which provides a highly robust and low-delay feedback signal for the ultrasonic probe to maintain a constant pressure on the tissue surface, and lays the foundation for subsequent accurate force control.

B. Cascade force/position hybrid control

In an automated ultrasound scanning robot, the end-effector is tasked with both precise position control along a predetermined trajectory and force control when the probe is in contact with the human body. In order to achieve the synergy between the two, we decompose the six-dimensional task space into a position subspace and a force subspace parallel to the contact direction. Let's start by defining the choice matrix for the force/position separation:

$$S_f = \text{diag}(0, 0, 1, 0, 0, 0), \quad S_p = I - S_f \quad (5)$$

where the third diagonal element of S_f is 1, which indicates that the Z-axis direction of the Cartesian space is selected as the force-controlled subspace; S_p is the position-controlled subspace composed of the remaining degrees of freedom. This decomposition satisfies orthogonality:

$$S_f + S_p = I, \quad S_f^T S_p = 0 \quad (6)$$

The projection mechanism allows for accurate force tracking in the Z-axis and position/attitude control in the XY axis and rotational degrees of freedom, effectively avoiding force-motion coupling problems.

1) tangential direction

A typical Cartesian impedance model is used in the tangential direction, the joint torque input to maintain the tangential impedance of the scan can be defined as:

$$\tau_{\text{task}}^{xy} = J(q)^T (S_p \cdot (-K_c(x - x_d) - D_c(\dot{x} - \dot{x}_d))) \quad (7)$$

where $J(q)^T$ is the Jacobian matrix of the end effector, x and $\dot{x}$ represent the current end position and velocity extracted from the robot state, x_d and $\dot{x}_d$ are the desired position and velocity. K_c and D_c are the position stiffness matrix and damping matrix, respectively. By dynamically adjusting the K_c , the system enables smooth and safe motion control in different environments.

2) normal direction

A cascade control structure with impedance feedforward to compensate for PID feedback in the normal direction. First, The dynamic properties of the second-order system consisting of mass-spring-damping embodied at the end of the robot of the ultrasound scan are as follows:

$$M_e \ddot{e}_z + D_e \dot{e}_z + K_e e_z = f_{\text{imp}} \quad (8)$$

The model depicts the effects of inertia, damping, and elasticity at the end in the direction of contact. where f_{imp} is the impedance compensation force feedback output, where e_z , $\dot{e}_z$, $\ddot{e}_z$ are the deviations of the robot position, velocity, acceleration and the expected value. M_e , K_e and D_e represent mass, stiffness, and damping matrices, respectively. The second-order model is approximated under the discrete control period, and the impedance compensation output is obtained:

$$f_{\text{imp}} = \left(\frac{M_e}{\alpha} - 1 \right) f_{\text{ext}} - \frac{M_e}{\alpha} (D_e \dot{e}_z + K_e e_z) \quad (9)$$

where f_{ext} is the external measured force, and α is the impedance scaling factor, and adjusting its value can change the system compliance. To further improve the force tracking performance, PID feedback control is introduced on top of the impedance output, and its joint torque input can be defined as:

$$\tau_F^z = J^T \left(S_f \cdot \left(K_p e_f + K_i \int_0^t e_f(\tau) d\tau + K_d \frac{d}{dt} e_f \right) \right) \quad (10)$$

where e_f is the force error, defined as $e_f = f_{\text{des}} + f_{\text{imp}} - f_{\text{ext}}$, f_{des} is the expected force. K_p , K_i and K_d are proportional, integral, and derivative gains, respectively, which are used to adjust the dynamic response of the PID controller.

C. Null space motion optimization

In robot motion control, null-space motion control is one of the core methods for redundant DOF systems. When the number of joints of the robotic arm exceeds the degrees of freedom required for the end-effector task, the system exhibits kinematic redundancy. The force control of most ultrasonic robots lacks the control of the null space motion of redundant robots, and this paper integrates null space motion to optimize the force control. The null space control of the 7-degree-of-freedom Franka manipulator adopts a rigorous mathematical

framework based on projection operators. The null space joint torque input can be defined as:

$$\tau_{\text{null}} = \left(I - J(q)^T \left(J(q)^T \right)^+ \right) \left(K_{\text{null}} (q_d - q) - D_{\text{null}} \dot{q} \right) \quad (11)$$

where $\left(J(q)^T \right)^+$ represents the pseudo-inverse, and the

null space projection matrix $N = \left(I - J^T \left(J^T \right)^+ \right)$ ensures that the control moment is strictly located in the zero-space, so as not to affect the execution of the main task. The stiffness term $K_{\text{null}} (q_d - q)$ adjusts the feedback intensity of the joint shape error $(q_d - q)$ through the scalar gain parameter K_{null} , and realizes the real-time dynamic adjustment by using the dynamic adjustment mechanism through the ROS external interface. In order to prevent the sudden change of torque in singular position, the joint error limiting mechanism was adopted, and the angle of each joint in null space was strictly confined to the $[-0.3, 0.3]$ rad interval. A semicritical damping configuration is used $D_{\text{null}} = 2\sqrt{K_{\text{null}}}$ to speed up convergence and maintain stability. Under the premise of not interfering with the main task, the module realizes the reasonable configuration and optimization of redundant degrees of freedom.

III. EXPERIMENTAL VALIDATION

To evaluate the efficacy of the proposed filter design and force control strategy for the robotic arm, a comprehensive set of experiments was conducted. Dynamic median filter compensation was first assessed by comparing raw and filtered force signals. Then, null-space optimization experiments analyzed joint fluctuation stability and safety performance with and without zero - space optimization. We also compared traditional PID control in abdominal phantom force tracking and tested shedding safety to assess system response during accidental robotic arm detachment. Lastly, high - force tracking performance was examined across plane, surface, and medical phantom environments.

A. Kinetic compensation median filter experiment

In order to verify the effectiveness of the designed filter on the smooth processing and noise cancellation of force signals. To test the designed filter, we carried out the following experiments: First, the robotic arm was fixed at a specific position in space so that its end was stationary. In the experiment, the operator abruptly presses or releases the end of the robotic arm with his hand to simulate the external disturbance forces that may be encountered in actual operation. This operation produces representative force variations, including abrupt changes in force and possible accompanying noise interference, providing a realistic test scenario for evaluating the filter's performance. During the experiment, we applied a sliding window median filter to the raw data, where the window length N was set to 99, to verify the effectiveness of the filter's smoothing and denoising ability on the force signal.

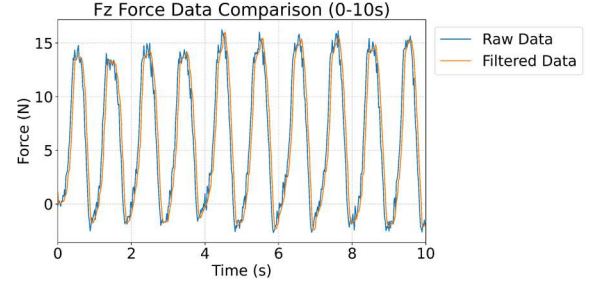


Fig. 2. Before and after comparison of the Fz force signal filter

The sampling frequency was set to 100 Hz and the recording duration was 10 s. Fig. 2 shows the Fz before and after optimization. The standard deviation of the original signal is 6.33N, and there is a clear pulse peak. With filter-optimized data, the standard deviation is reduced to 6.23N and the noise is reduced by 1.3%, while preserving the authenticity of the data and making it smoother. Experimental results show that the filter can effectively suppress the measurement noise and reduce the force control error, so as to improve the force control accuracy and stability of the manipulator in ultrasonic tasks, and verify the effectiveness of the proposed filtering method.

B. Null space motion optimization effectiveness experiment

In order to verify the effectiveness of the zero-space motion optimization control method, the 7-degree-of-freedom manipulator ultrasonic robot was used as the experimental platform, and the motion trajectory of the end probe of the experimental control robot was used as a linear motion along the tangential y-axis of the scanning inspection. The y-axis velocity is 0.02 m/s and the running time is 10 s. The initial joint angles of the ultrasound robot were $[-0.14, -0.18, -0.32, -2.41, -0.10, 2.23, 0.34]$, ensuring that the robotic arm was operated in the same position for both experiments.

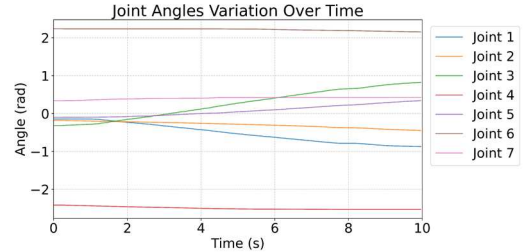


Fig. 3. No null space optimization

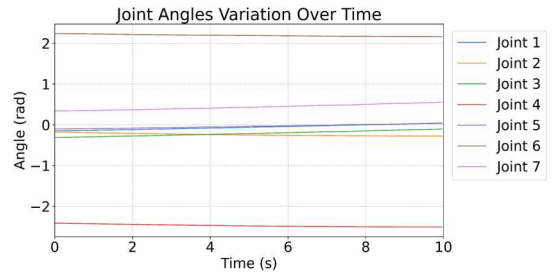


Fig. 4. Null space optimization

Fig. 3 and Fig. 4 illustrate the change of joint angles over time before and after optimization, respectively. The comparison shows that the fluctuation amplitude of the joint angle is significantly reduced and the curve is more stable after optimization, which indicates that the null space motion optimization effectively inhibits the unnecessary fluctuation

of the joint angle, improves the stability and accuracy of the robotic arm movement, and verifies the effectiveness of the proposed method.

C. Abdominal phantom force tracking experiment

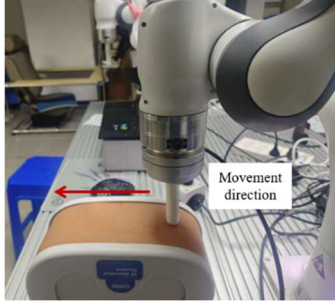


Fig. 5. Experimental process

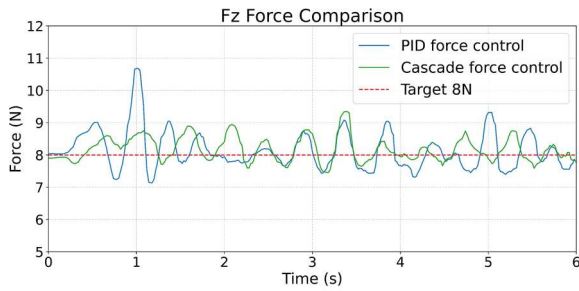


Fig. 6. Fz force tracking data

In order to verify the advantages of the cascaded force control method proposed in this paper in terms of stability and safety, we conducted force tracking control experiments in the abdominal phantom force contact environment, and compared it with the traditional PID force control method. To ensure the reliability and reproducibility of the results, 10 experiments were performed independently for each control strategy. In the experiments, both methods used the same PID controller basic parameters (proportional gain $K_p = 1$, integral gain $K_i = 2$, and differential gain $K_d = 0.01$). Traditional PID force control methods only use this PID parameter set. The cascaded force control method proposed in this paper adds an impedance compensation control strategy on the basis of the PID force control, and its core parameters are as follows: mass term $M_c = 3$ kg, damping term $D_c = 31.26$ Ns/m, stiffness term $K_c = 50$ N/m, and scaling coefficient $\alpha = 3$. The experimental process is shown in Fig. 5, the end probe of the robot is controlled to move in a straight line along the tangential x-axis of the soft medical phantom scan, with a velocity of 0.02 m/s and a duration of 6 s, and the ultrasound probe maintains a contact force of 8 N (as shown in the red line in Fig. 6).

Fig. 6 shows a comparison of the normal contact force (F_z) traces for one of the typical experiments (blue line for PID force control and green line for cascaded force control). The results show that compared with the traditional PID control (standard error: 0.53 N, maximum contact force: 10.68 N), the cascaded force control method proposed in this paper significantly improves the performance (standard error: 0.39 N, maximum contact force: 9.35 N). Further analysis of the average data of 10 experiments shows that the mean standard error of the cascade control is 0.40 N, the average error of the maximum contact force is 9.48 N, while the mean standard

error of the PID control is 0.53 N, and the average error of the maximum contact force is 9.89 N. Cascaded force control reduces the standard error of the average force by 24.53%, which means less fluctuation of force and better stability; At the same time, the maximum contact force is reduced by an average of 4.15%, which also proves that it has a higher level of interaction safety.

In addition, in order to verify the effectiveness of the method in dealing with extreme interactive environments such as accidental probe shedding, we also conducted simulated shedding experiments. During the experiment, the robotic arm moves along the x-axis of the fixed plane until it falls off the detachment platform. The experimental results show that the impedance compensation increases rapidly when the robot arm is detached from the scanned object (as shown in Fig. 7), and the robot arm tends to stabilize after dropping a short distance in the normal direction (as shown in Fig. 8). The overall experiment shows that the proposed method not only performs well in terms of compliance and stability, but also has good shedding safety performance, which can effectively ensure the safety of the robotic arm and the surrounding environment in unexpected situations.

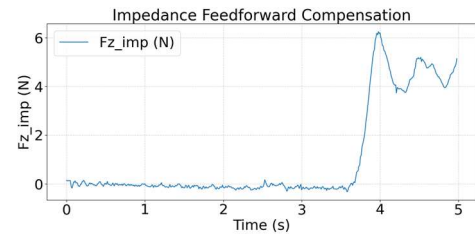


Fig. 7. Impedance compensation force

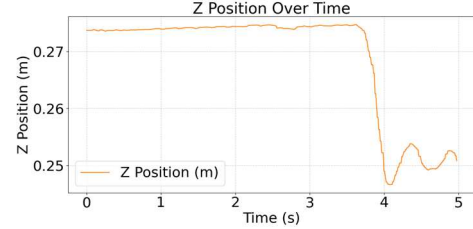
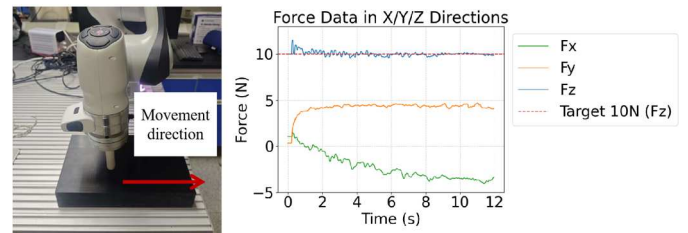
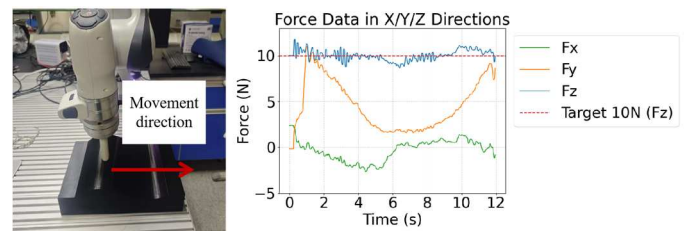


Fig. 8. Normal displacement deviation

D. Cascade force/position hybrid control experiments



(a)



(b)

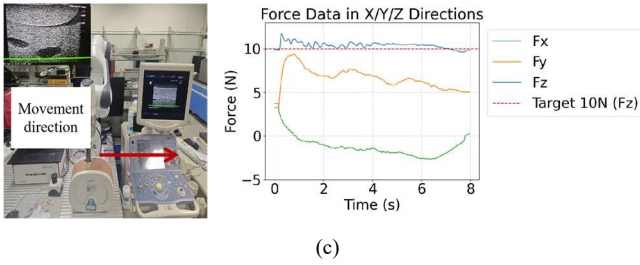


Fig. 9. Ultrasound robot hybrid force control experiment: (a) plane force control, (b) curved force control, and (c) abdominal phantom force control.

In order to verify the effectiveness and reliability of the force control algorithm of the ultrasound robot cascade force/position hybrid controller designed in this paper based on redundancy optimization in the unknown or complex environment in contact with human tissue, force tracking control experiments were carried out in three different force contact environments: plane, curved surface and medical abdominal phantom. The experimental parameters are consistent with the above cascade force stability safety and shedding safety performance experiments. During the experiment, the end of the control ultrasound robot moves in the tangential direction of the y-axis of the ultrasound scan, and the ultrasound probe maintains the required normal contact force F_z (as shown by the blue line in Fig. 9) for a distance. In the plane experiment and the surface experiment, the ultrasonic robot moves in the tangential direction of the y-axis at a speed of 0.02 m/s and the expected force F_z (as shown in the red line in Fig. 9) is set to 10N. In the medical abdominal phantom experiment, the constant force tracking was carried out with an ultrasound probe, the robot moved at a speed of 0.02 m/s, and the expected force F_z was set to 10 N.

The experimental results are shown in Fig. 9, where Fig. 9 (a), Fig. 9 (b) and Fig. 9 (c) are the force tracking control experimental scenarios and robot end force observation results of the ultrasonic robot on the plane experimental platform, the curved experimental platform and the medical abdominal phantom, respectively. The average tracking error of the scanning normal contact force on the plane experimental platform is 0.16 N, the maximum contact force error is 1.50 N, the average tracking error of the force on the curved surface experimental platform is 0.44 N, the maximum contact force error is 1.80 N, the average tracking error of the force on the abdominal phantom is 0.46 N, and the maximum contact force error is 1.73 N. Fig. 9 (c) is the upper left image of the gray scar group obtained by the ultrasound robot on the medical abdominal phantom.

IV. CONCLUSION

In this paper, we propose a cascaded force/position hybrid control method for ultrasonic robots based on redundancy optimization. In the tangential direction of the ultrasonic robot scanning, Cartesian impedance control is adopted. In the scanning mode, a cascade control strategy that fuses impedance feedforward and PID feedback ensures good compliance while maintaining a stable contact force. In the cascade force stability safety experiment, compared with the traditional PID control, the cascaded force control reduced the standard error of the contact force by 24.53% and the maximum contact force by 4.15%, which effectively

improved the stability and safety of force tracking. The shedding safety experiment also proves that the method can quickly respond to the accidental shedding of the robotic arm and effectively ensure the safety of the system. In addition, a dynamic compensation median filter is designed, which can effectively suppress the impulse noise while retaining the real force mutation. At the same time, by introducing the null-position space optimization into the force/impedance hybrid control framework, the redundant degrees of freedom are effectively used to suppress the joint angle fluctuation. Finally, experiments were carried out in three contact environments: plane, curved surface and medical phantom, and the cascaded force/position hybrid control method showed good robustness.

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